

Fig. 3: Front Panel of LabVIEW

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

drop it on Front Panel and arrange in the form of an actual 7-segment LED display and name each as individual segment a, b, c, d, e, f and g, respectively, as shown in Fig. 3.

4. Go to Block Diagram workspace (Fig. 4). From Menu, select Programming→Boolean→NOT gate; drag and drop it on the workspace for each input as shown in Fig. 4.

5. From Menu, select Programming→Numeric→Compound Arithmetic; drag and drop it on the workspace. Right-click on Compound Arithmetic, select Change Mode and choose logic gate (OR or AND) operation, as needed. Compound Arithmetic

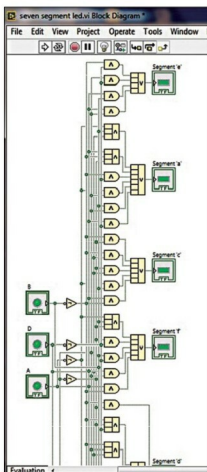


Fig. 4: Block diagram of LabVIEW for the display

tic is used to carry out logical and arithmetic operations for more than two inputs.

6. Similarly, select suitable logic gates for the corresponding Boolean expression for each segment as given in Table II.

7. Connect each output of the Boolean expression to each of the indicator (segment) as shown in Fig. 4.

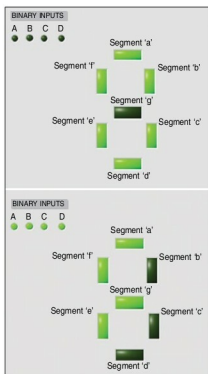


Fig. 5: Decimal/hexadecimal display for 0 and F

8. Return to Front Panel. Save VI and press Run Continuously to check the operation of 7-segment display giving different binary inputs and also check outputs from 0 to f.

Checking the output

Fig. 5 shows the result of a 7-segment LED (decimal and hexadecimal) display for 0 and F. ●



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